Fiscal Year:	FY 2023	**Task Last Updated:**	FY 11/07/2022
PI Name:	Ramasubramanian, Anand Ph.D.		
Project Title:	Thrombosis in Microgravity		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	(1) Cell Culture		
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PI Organization Type:	UNIVERSITY	**Phone:**	408-924-3922
Organization Name:	San Jose State University Research Foundation		
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City:	San Jose	**State:**	CA
Zip Code:	95192-1000	**Congressional District:**	19
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2018 Space Biology (ROSBio) NNH18ZTT001N-FG2. App D: Flight and Ground Space Biology Research
Start Date:	12/01/2020	**End Date:**	11/30/2023
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA ARC
Contact Monitor:	Griko, Yuri	**Contact Phone:**	650-604-0519
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 11/30/2023 per NSSC information (Ed., 11/2/22) NOTE: End date changed to 11/30/2022 per NSSC information; original end date was 11/30/2021 (Ed. 10/26/21)		
Key Personnel Changes/Previous PI:	None		
COI Name (Institution):	Lee, Sang-Joon Ph.D. (San Jose State University Research Foundation) Lee, Wendy Ph.D. (San Jose State University Research Foundation)		
Grant/Contract No.:	80NSSC21K0272		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Dramatic changes in cardiovascular physiology during spaceflight are well-documented: redistribution of fluids from lower to upper part of the body, a remarkable decrease in blood volume, and an increase in stroke volume and cardiac output. The headward fluid shift due to weightlessness results in sustained intracranial pressure. In January 2020, the New England Journal of Medicine (NEJM) report detailed the first ever occlusive thrombotic event in the internal jugular vein (IJV) of an astronaut in the International Space Station, and a 2019 JAMA (Journal of the American Medical Association) study reported that structural changes in the internal jugular vein of astronauts, and fluid congestion due to headward fluid shift as the newly discovered risk factors of stroke during space travel. These studies reveal gaps in our understanding of circulatory and hemostatic physiology during space travel. Our objective of this work is to understand the effect of microgravity and headward fluid shift on the risk of venous thrombosis. We will test the hypothesis that microgravity modulates the risk of cerebral venous thrombosis through effects on endothelial activation and local transport dynamics. Using a scale-down microfluidic endothelialized model of the vasculature subjected to microgravity in a random positioning machine (RPM), we will test the hypothesis through the following specific aims: (i) evaluate the combined effect of microgravity and venous flow on endothelial transcriptome and miRNA-ome; and (ii) evaluate the effect of microgravity on venous thrombosis. We expect that our investigative team with expertise in vascular mechanobiology, microfluidics, and system biology will provide insights into poorly understood aspects of microgravity on cardiovascular physiology, which have been mentioned as an area of Space Biology research emphases, and also as high priority in the midterm assessment of the Decadal report. Further, in our opinion, the recent incidents raise the risk of thrombosis as a health concern in astronauts. We believe that our innovative ground analog will generate testable hypothesis, and new, high-value tools that can be translated to full scale spaceflight investigations on hemostasis and thrombosis. References: Auñón-Chancellor SM, Pattarini JM, Moll S, Sargsyan A. Venous Thrombosis during Spaceflight. N Engl J Med. 2020 Jan 2;382(1):89-90. http://doi.org/ ; PMID: 31893522 . Marshall-Goebel K, Laurie SS, Alferova IV, Arbeille P, Auñón-Chancellor SM, Ebert DJ, Lee SMC, Macias BR, Martin DS, Pattarini JM, Ploutz-Snyder R, Ribeiro LC, Tarver WJ, Dulchavsky SA, Hargens AR, Stenger MB. Assessment of Jugular Venous Blood Flow Stasis and Thrombosis During Spaceflight. JAMA Netw Open. 2019 Nov 1;2(11):e1915011. http://doi.org/ ; PMID: 31722025 ; PMCID: PMC6902784 .
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Understanding the effects of hydrodynamic and gravitational forces on the growth and function of endothelial cells, which line the blood vessels, will provide deeper insights into both health and disease. Fundamental insights into how forces may help maintain cardiovascular health, and how perturbations may result in thrombosis, will be attained.
Task Progress:	• We have been working on two convergent projects: one focused on endothelial cell culture in microgravity and subjected to physiological shear stresses, and the other focused on the effects of blood flow on venous thrombosis. • We have developed and characterized an in-house closed-loop flow system for perfusion culture for endothelial cells under normal and microgravitational conditions. The phenotypic and genotypic characterization of culture cells will allow us to evaluate the combined effects of two types of forces. • We have characterized the spatial distribution of red blood cells (RBCs) in a microfluidic model of venous valve. We used high speed microscopy and image analysis, particle image velocimetry, and 3D computational fluid dynamics simulations to explore the interaction between fluid flow and RBC distribution. The results show that the local hematocrit in the valve pockets is spatially heterogeneous and is significantly different from the feed hematocrit. Above a threshold shear rate, the inertial separation of streamlines and lift forces contribute to an uneven distribution of RBCs in the vortices, the entrapment of RBCs in the vortices, and non-monotonic wall shear stresses in the valve pockets. Our experimental and computational characterization provides insights into the complex interactions between fluid flow, RBC distribution, and wall shear rates in venous valve mimics, which is of relevance to understanding the pathophysiology of thrombosis. • Bringing together these two approaches and results will enable us to understand the biophysical origins of venous thrombosis observed in certain astronauts aboard the International Space Station (ISS).
Bibliography Type:	Description: (Last Updated: 11/21/2025)
Articles in Peer-reviewed Journals	Sanchez ZAC, Vijayananda V, Virassammy DM, Rosenfeld L, Ramasubramanian AK. "The interaction of vortical flows with red cells in venous valve mimics." Biomicrofluidics. 2022 Mar 3;16(2):024103. ttps://doi.org/10.1063/5.0078337 ; PMID: 35282036 PMCID: PMC8896891 , Mar-2022